COMMON QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7/1	(a)	(i)		award (1) for any of following • leave to crystallise / evaporate / dry naturally • leave to dry for a few days / until next lesson • leave to dry in a warm place / on window sill / on radiator must have a 'process' and the idea that it happens over a period of time **OR** in a warm place neutral answer – leave to dry	1			1		1
		(ii)		no fizzing / bubbles / effervescence (with oxide) (1) because no carbon dioxide produced (1) alternative answer black powder (rather than green) would be left in the beaker when all the acid has reacted (1) because copper(II) oxide is black (not green) (1)			2	2		2
		(iii)		$CuSO_4$ + H_2O award (1) for each correct product		2		2		

<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="4">Marks available</th><th></th><th></th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>
<table>
<tr><td>Part of the energy profile</td><td>Letter</td></tr>
<tr><td>energy change for the reaction</td><td>C</td></tr>
<tr><td>energy of the reactants</td><td>A</td></tr>
<tr><td>activation energy of the reaction</td><td>B</td></tr>
</table>

award (2) for all three correct
award (1) for any one correct</td><td>2</td><td></td><td></td><td>2</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>the (minimum) energy required for a reaction to happen / start

accept ‘the <u>minimum</u> energy required to activate the reaction’
neutral answer – the energy required to activate the reaction</td><td>1</td><td></td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>award (1) for any of following

• the energy of the products is lower than the energy of the reactants
• the product line is below the reactant line / E is below A
• energy given out is greater than energy taken in / D is greater than B
• lower energy at the end than at the beginning

neutral answer – negative energy change</td><td></td><td>1</td><td></td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 7/1 total</td><td>4</td><td>3</td><td>2</td><td>9</td><td>0</td><td>3</td></tr>
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